I carefully analyzed your uploaded document . It describes a **Corporate Employee Transport Management System** used by companies to manage employee cab transportation, automate trip planning, provide live tracking, safety alerts, and generate billing reports.

Below is a **client-ready documentation** with:

- System architecture
- Tech stack
- UI modules
- Database design
- Features
- Flow diagrams
- Development phases

Corporate Employee Transport Management System

Complete Software Architecture & Development Plan

```
                    +----------------------+
                    |      Super Admin     |
                    |   (Platform Owner)   |
                    +----------+-----------+
                               |
                               |
                        Web Admin Panel
                               |
                               |
             +-----------------+------------------+
             |                                    |
     Company Admin Dashboard              Reporting & Billing
             |
             |
    +--------+--------+--------+--------+
    |                 |                 |
Employee App      Driver App        Admin Map
(Mobile)          (Mobile)        (Live Tracking)
    |                 |                 |
    +--------+--------+--------+--------+
                      |
                 API Backend Server
                      |
        +------------+------------+
        |                         |
   Real-Time Services        Background Workers
   (WebSocket / Redis)       (Celery + Redis)
        |                         |
        +------------+------------+
                      |
                 PostgreSQL
                  Database
                      |
                 PostGIS Extension
                      |
              Cloud Infrastructure
           (AWS / Azure / DigitalOcean)
```

1. Project Overview

The **Corporate Employee Transport Management System (ETMS)** is a cloud-based platform that automates the entire employee transportation process for companies.

It replaces manual systems like Excel sheets, driver calls, and manual route planning.

According to the uploaded document, the system manages:

- Employee pickup & drop
- Driver assignment
- Vehicle tracking
- Route planning
- SOS safety alerts
- Reporting & billing

Core Objective

Provide a **secure, automated, real-time transport management system** for organizations with employee transportation needs.

---

2. Users of the System

The document shows **three primary user types**.

User	Platform	Purpose
Super Admin	Web Panel	Manage companies
Company Admin	Web Panel	Manage transport
Driver	Mobile App	Execute trips
Employee	Mobile App	Track cab

---

3. High Level System Architecture

```
+----------------------+
|   Super Admin        |
| (Platform Owner)     |
+----------+-----------+
           |
           |
      Web Admin Panel
           |
           |
+--------------+--------------+
```

```
        |                       |
    Company Transport Team        Reporting System
        |                       |
        |
   +-------+-------+
   |             |
Driver Mobile App   Employee Mobile App
   |             |
   |             |
   +-------+-------+
        |
     API Backend Server
        |
     Real-Time Services
        |
      Database
```

---

4. Recommended Technology Stack

Frontend (Admin Dashboard)

Best choice:

Next.js (React)
Tailwind CSS
TypeScript
Redux Toolkit
React Query
Chart.js

Why:

- Fast dashboard UI
- SEO support
- Real-time updates

Mobile Apps

Two apps:

1. Driver App

2. Employee App

Best option:

React Native
or
Flutter

Features needed:

- GPS tracking
- Push notifications
- Maps navigation
- SOS alerts

Backend

Recommended:

Python or Node
Django + Django REST Framework
Celery
Redis

Why:

- Secure
- Scalable
- Easy API development

Database

Recommended:

PostgreSQL

With extensions:

PostGIS

Reason:

Location queries for **map-based pickup points**.

Real Time Tracking

Use:

WebSockets
Socket.IO
or
Firebase Realtime DB

Purpose:

- Live vehicle location
- Live admin dashboard

Maps Integration

Google Maps API
Mapbox

Features:

- Route optimization
- Navigation
- Distance calculation

Notifications

Firebase Cloud Messaging
Twilio SMS
Email (SendGrid)

Used for:

- Cab arrival alert
- SOS alerts
- Admin notifications

---

5. Core System Modules

1. Super Admin Panel

Functions:

- Create company accounts
- Manage subscriptions
- Monitor usage
- Platform analytics

Features

Company Management
Billing Plans
System Logs
Admin Controls

---

2. Company Admin Dashboard

Main operational panel.

Modules:

Dashboard
Employee Management
Driver Management
Vehicle Management
Shift Management
Trip Planning
Route Optimization
Live Tracking
Reports & Billing

---

6. System Workflow (Full Process)

Based on the document flow.

System Flow Diagram

Company Onboarding

```
        |
        v
Add Employees + Drivers + Vehicles
        |
        v
Assign Shift Timing
        |
        v
Generate Daily Roster
        |
        v
Auto Trip Creation
        |
        v
Driver Accepts Trip
        |
        v
Trip Starts (Live Tracking)
        |
        v
Employee Pickup
        |
        v
Employee Drop
        |
        v
Trip Completed
        |
        v
Reports Generated
```

This matches the **step-by-step system process described in the document**.

---

7. Trip Generation Algorithm

System automatically groups employees.

Logic:

1 Fetch employees by shift
2 Sort by location
3 Cluster nearby employees
4 Assign vehicle capacity
5 Assign available driver

6 Generate optimal route

Example

Shift: 9:00 AM
Employees: 32

Vehicle capacity: 6

Trips generated = 6

---

8. Live Tracking Architecture

```
Driver GPS
    |
    v
Mobile App
    |
    v
Location API
    |
    v
WebSocket Server
    |
    v
Admin Dashboard Map
    |
    v
Employee App Map
```

Update frequency:

Every 5 seconds

---

9. Safety System (SOS)

Important feature from the document.

When driver presses SOS:

```
Driver App
  |
  v
SOS API Trigger
  |
  v
Alert System
  |
  +---- SMS to Manager
  |
  +---- Admin Dashboard Alert
  |
  +---- Save GPS Location
```

This provides **emergency safety monitoring**.

---

10. Database Design

Core Tables

companies
users
employees
drivers
vehicles
departments
shifts
routes
trips
trip_stops
driver_locations
alerts
reports
invoices

Example ER Diagram

```
Company
  |
  |---- Employees
  |
  |---- Drivers
```

```
|
|---- Vehicles
|
|---- Shifts
|
|---- Trips
        |
        |---- Trip Stops
        |
        |---- Driver
        |
        |---- Vehicle
```

---

11. Admin Dashboard Screens

Main screens:

Dashboard
Employee List
Driver List
Vehicle List
Trip Planner
Live Map
Shift Manager
Reports
Billing

---

12. Mobile App Features

Driver App

Features:

Login
Today's Trips
Route Navigation
Start Trip
Pickup Confirmation
Drop Confirmation
SOS Button

Trip History

Employee App

Features:

Login
Trip Notifications
Driver Details
Vehicle Details
Live Map Tracking
ETA
Trip History
Emergency Contact

---

13. Reporting & Analytics

At end of month system generates:

- Total trips
- Total distance
- Driver performance
- Delay reports
- Cost per employee

These are mentioned in the final step of the document.

---

14. Security Design

Security layers:

JWT Authentication
Role Based Access Control
HTTPS
Encrypted Passwords
Audit Logs
API Rate Limiting

---

15. Deployment Architecture

Cloud setup:

```
Frontend (Next.js)
    |
    v
CDN (Cloudflare)
    |
    v
API Server (Django)
    |
    v
Redis + Celery
    |
    v
PostgreSQL Database
```

Hosting options

AWS
DigitalOcean
Azure

---

16. Estimated Development Timeline

Phase
Planning
UI/UX Design
Backend Development
Mobile Apps
Integration
Testing
Deployment

Total

4 weeks

17

---

18. Project Folder Structure

Backend

```
transport-system
  |
  |-- users
  |-- employees
  |-- drivers
  |-- vehicles
  |-- trips
  |-- routes
  |-- notifications
  |-- billing
```

---

19. UI/UX Design Concept

Admin Dashboard

Left Sidebar
Dashboard
Employees
Drivers
Vehicles
Trips
Reports
Billing

Map centered interface.

---

20. This system will provide:

✓ Automated employee trip planning
✓ Real-time vehicle tracking
✓ Driver mobile operations
✓ Employee trip tracking
✓ Safety alerts (SOS)
✓ Detailed analytics & billing

---

21. Configurable Transport Rules Engine

Overview

The platform includes a **Configurable Transport Rules Engine** that allows organizations to define operational constraints for employee transportation.

This system ensures that **transport planning, trip generation, and route optimization follow company-specific policies**.

Each organization (tenant) can configure their own rules independently.

This is essential in multi-tenant systems because **different companies have different transport policies**.

Transport Rules Engine Architecture

Company Admin Dashboard

|

v

Transport Rules Configuration

|

v

Rules Engine

|

v

Trip Generation Service

|

v

Route Optimization Engine

Rules are applied **before trips are generated**.

Configurable Transport Rules

The following operational rules can be configured per organization.

Rule Category	Description
Pickup Time Window	Define how early employees can be picked before shift
Female Safety Rule	Female employee cannot be last drop in night shift
Vehicle Capacity Utilization	Minimum and maximum occupancy rules
Maximum Route Duration	Limit total travel time
Driver Shift Limits	Restrict driver working hours

Maximum Pickup Distance	Limit distance deviation from route
Last Pickup Rule	Prevent unsafe pickup sequences
Night Shift Safety	Mandatory security escort policies

Example Rule Configuration

Example:

Shift Start Time: 9:00 AM

Pickup Window: 30 minutes before shift

Vehicle Capacity: 6

Minimum Utilization: 4 employees

Max Route Duration: 60 minutes

Driver Max Working Hours: 10 hours

Female Safety Policy Example

For night shifts:

Rule:

Female employee cannot be the last drop.

Solution:

System reorders route automatically.

OR

Assign security escort.

Flow:

Trip Generation

|

Check Safety Rules

|

If violation found

|

Route Adjusted

Rule Storage Model (Database)

Example tables:

transport_rules

rule_types

organization_rules

rule_conditions

rule_values

Example structure:

id	organization_id	rule_type	rule_value

1	201	pickup_window	30 minutes
2	201	max_route_duration	60 minutes
3	201	female_last_drop_allowed	false

Rule Processing Flow

Trip Request Generated

|

v

Load Organization Rules

|

v

Validate Constraints

|

v

Apply Route Optimization

|

v

Generate Final Trip Plan

Example Trip Generation With Rules

Input:

Employees = 25

Shift = 9 AM

Vehicles = 5

Capacity = 6

Constraints applied:

Pickup window = 30 minutes

Max route duration = 60 minutes

Female last drop = not allowed

Result:

Trips Generated = 5

Routes Optimized

Rules satisfied

Rules Engine Technology Implementation

Recommended implementation:

Backend: Django Rules Engine

Processing: Celery Workers

Caching: Redis

Optimization: Python OR-Tools

Possible optimization libraries:

Google OR Tools

OSRM

GraphHopper

These allow **vehicle routing with constraints**.

Admin UI for Rule Configuration

Transport managers can configure rules through the dashboard.

Example UI:

```
Transport Settings
    |
    |---- Pickup Window
    |---- Female Safety Rule
    |---- Route Duration Limit
    |---- Vehicle Capacity Rules
    |---- Driver Work Hours
```

Each organization can define its own policies.

Benefits of Rules Engine

The rules engine ensures:

✓ Safety compliance
✓ Policy enforcement
✓ Automated route planning

✓ Flexible configuration per tenant
✓ Reduced manual intervention

Example Rule Engine Diagram

Employee Data

|

v

Shift Assignment

|

v

Rules Engine

|

v

Route Optimization

|

v

Trip Generation

Additional Update (Recommended for Client)

Also mention in documentation:

Enterprise Transport Intelligence

The platform includes analytics for:

- Vehicle utilization
- Route efficiency
- Cost per employee
- Fleet optimization
- Driver performance

Update for Role Permission Matrix

The permission matrix image you attached represents **Role Based Access Control (RBAC)**.

You should add a section:

Role-Based Access Control

Roles:

Super Admin

Tenant Admin

HR

Transport Manager

Ground Staff

Finance

Driver

Employee

Permissions are controlled using **RBAC policies stored in database**.

Final Result

After adding this section, your documentation will include:

1️⃣ Architecture
2️⃣ Tech stack

3️⃣ Workflow
4️⃣ Rules engine
5️⃣ RBAC system
6️⃣ Scalability design
7️⃣ Analytics

---

1. Multi-Tenant Architecture Confirmation

The platform is designed using a **multi-tenant SaaS architecture**, allowing multiple organizations to operate independently within the same system.

Each organization is treated as a **separate tenant**, and all tenant-specific data is associated with a unique `tenant_id`.

Tenant data isolation is implemented through:

- tenant-aware middleware
- tenant-scoped database queries
- role-based access controls

This ensures that users can only access data belonging to their own organization while maintaining a shared infrastructure environment.

---

2. CI/CD and DevOps Workflow

The development lifecycle follows a **Continuous Integration and Continuous Deployment (CI/CD) pipeline** to ensure stable and reliable software releases.

Development Workflow

Developer Commit

↓

Code Pushed to Git Repository

↓

Automated CI Pipeline Triggered

↓

Code Validation and Testing

↓

Build Artifacts Generated

↓

Deployment Pipeline Executed

Automated checks include:

- code linting
- unit testing
- dependency validation
- build verification

This workflow ensures consistent code quality and rapid deployment of new features.

---

3. Git Repository Ownership

All project source code will be maintained in a centralized **Git-based version control system**.

The repository will be created under the **client's organization account** from the start of the project to ensure full ownership of the product.

Supported platforms include:

- GitHub
- GitLab
- Bitbucket

A structured branching strategy will be followed:

main → production releases

develop → integration branch

feature branches → new development

hotfix branches → urgent fixes

This ensures controlled development workflows and transparent version tracking.

Additional Client Clarifications & System Design Updates

This section includes additional technical clarifications requested by the client to ensure the platform supports **enterprise-scale SaaS transport operations with strong security, automation, and operational flexibility**.

The sections below extend the existing architecture documentation without repeating previously described system components.

---

1. Enhanced Multi-Tenant Security Architecture

The platform already supports multi-tenant SaaS architecture using tenant-based logical isolation.
To further strengthen security and ensure strict separation between organizations, additional security mechanisms are implemented.

Middleware-Based Tenant Resolution

Tenant identification is handled through a dedicated middleware layer.

Tenant Resolution Flow

User Login

↓

Tenant Resolution Middleware

↓

Tenant Context Applied

↓

Database Queries Filtered by tenant_id

↓

Tenant-Specific Data Returned

The middleware automatically identifies the organization (tenant) associated with each request and applies the appropriate tenant context to all backend operations.

This ensures that:

- all API queries are automatically scoped to the correct tenant
- cross-tenant data access is prevented
- tenant configurations remain isolated

PostgreSQL Row-Level Security (RLS)

To provide **database-level tenant protection**, PostgreSQL Row-Level Security (RLS) can be enabled.

RLS ensures that database queries only return rows belonging to the authenticated tenant.

Example policy:

```
CREATE POLICY tenant_isolation

ON employees

USING (tenant_id = current_setting('app.current_tenant')::int);
```

Benefits:

- prevents cross-tenant data leakage
- enforces security at the database layer
- provides defense-in-depth protection

---

2. Advanced Route Optimization Engine

The transport platform includes an advanced route planning system designed to generate efficient employee transport routes.

Optimization Engine

The system uses the **Google OR-Tools Vehicle Routing Problem (VRP) solver** for route optimization.

OR-Tools enables advanced optimization with multiple operational constraints.

Optimization Constraints

The trip generation engine evaluates several operational parameters:

Parameter	Description
Vehicle Capacity	Maximum passengers per vehicle
Pickup Time Window	Allowed pickup range before shift
Shift Timing	Employee shift start time
Driver Availability	Driver work schedule
Female Safety Rule	Female employee cannot be last drop
Maximum Route Duration	Limit total travel time
Vehicle Utilization	Minimum occupancy threshold

These constraints ensure that generated routes comply with operational policies and safety requirements.

Constraint-Based Routing Strategy

The system does **not rely solely on Google Maps shortest path routing**.

Instead, the routing process follows:

Constraint-Based Optimization

Mapping services such as Google Maps or Mapbox are used for:

- travel distance calculations
- ETA estimation
- driver navigation

The final route sequence is generated by the VRP optimization engine.

---

3. Automated On-Call Transport Booking

The system supports automated processing of ad-hoc transport requests received from clients.

Transport managers can upload transport booking sheets directly into the platform.

Excel / CSV Import

Supported formats:

- Excel (.xlsx)
- CSV (.csv)

Typical booking sheet columns include:

Employee Name

Employee ID

Pickup Location

Shift Time

Route Number

Department

Automated Booking Processing

After upload, the system performs automated processing.

Import Booking Data

↓

Validate Employee Records

↓

Match Pickup Locations

↓

Group Employees

↓

Generate Trips Automatically

This removes the need for manual trip planning.

Employee Grouping Logic

Employees can be grouped using multiple criteria.

Grouping Method	Description
Route Number	Predefined company route

Pickup Location	GPS-based clustering
Department	Optional grouping rule
Shift Timing	Mandatory grouping parameter

This allows efficient trip generation while maintaining operational flexibility.

---

4. Advanced Transport Rules Configuration

The platform includes a configurable rules engine that allows organizations to define operational transport policies.

Each organization can configure its own rules independently.

Configurable Rule Types

Rule	Description
Pickup Time Window	Example: 20–40 minutes before shift
Female Safety Policy	Female employee cannot be last drop
Maximum Route Duration	Limit total travel time

Vehicle Capacity Limits	Minimum and maximum passengers
Driver Shift Duration	Maximum driver working hours
Maximum Pickup Distance	Allowed deviation from route

Rule Storage Model

Rules are stored per organization in the database.

Example structure:

transport_rules

---

id

organization_id

rule_type

rule_value

Example configuration:

organization_id = 201

pickup_window = 30 minutes

max_route_duration = 60 minutes

female_last_drop_allowed = false

This design allows flexible policy management across different companies.

---

5. Real-Time Tracking Architecture (Enhanced)

The platform supports real-time GPS monitoring of transport vehicles.

GPS Update Frequency

Driver mobile applications send location updates approximately every:

5 seconds

This provides near real-time trip visibility.

Real-Time Streaming Architecture

Driver Mobile App

↓

GPS Location API

↓

WebSocket Server

↓

Redis Pub/Sub

↓

Admin Dashboard Map

Employee Mobile App

Redis Pub/Sub ensures efficient distribution of live tracking updates across multiple dashboards and users.

Tracking Capacity

Typical deployment supports:

Parameter	Capacity
Vehicles per organization	100–500
Concurrent GPS updates	Thousands
Dashboard users	Multiple administrators

The system can scale horizontally for larger deployments.

---

6. Asynchronous Background Job Processing

Several heavy system operations are executed using asynchronous background processing.

Technology Stack

Celery Workers

Redis Queue

Background Tasks

Examples of asynchronous tasks:

- trip generation
- bulk employee imports
- analytics calculations
- notification delivery
- report generation

Task Processing Flow

User Request

↓

API Receives Request

↓

Task Added to Redis Queue

↓

Celery Worker Processes Task

↓

Results Stored in Database

↓

User Notification Sent

This architecture keeps the API responsive even during large processing workloads.

---

7. Enterprise Scalability Expectations

The platform architecture supports large enterprise transport deployments.

Typical Deployment Capacity

Component	Supported Scale
Organizations (tenants)	Hundreds
Employees per organization	5,000+
Vehicles tracked simultaneously	500+
Active trips	Thousands

Large Dataset Trip Generation

Example scenario:

Employees: 1000

Vehicles: 150

Multiple operational constraints

Trip generation typically completes within:

10–30 seconds

depending on route complexity.

---

8. Transport Analytics & Fleet Optimization

The platform provides advanced analytics dashboards to help organizations monitor and optimize transport operations.

Key Analytics Metrics

Metric	Description
Vehicle Utilization	Passenger occupancy rates
Route Efficiency	Travel time optimization
Transport Cost per Employee	Financial cost analysis
Fleet Optimization	Route performance insights
Driver Performance	Punctuality and safety metrics

These insights help organizations reduce operational costs and improve service efficiency.

---

9. Role-Based Access Control (RBAC)

The platform implements a comprehensive Role-Based Access Control system to manage permissions across different operational roles.

The following roles are supported:

- Super Admin
- Tenant Admin
- HR
- Transport Manager
- Ground Staff
- Finance

- Driver
- Employee

Role Permission Matrix

Module / Action	Super Admin	Tenant Admin	HR	Transport Manager	Ground Staff	Finance	Driver	Employee
Create Organization	✓	✕	✕	✕	✕	✕	✕	✕
Suspend Organization	✓	✕	✕	✕	✕	✕	✕	✕
View All Tenants	✓	✕	✕	✕	✕	✕	✕	✕
Manage Subscription Plan	✓	✕	✕	✕	✕	✕	✕	✕
Manage Organization Settings	✕	✓	✕	✕	✕	✕	✕	✕
Create Admin Users	✕	✓	✕	✕	✕	✕	✕	✕
Upload Employee Excel	✕	✓	✓	✕	✕	✕	✕	✕

Edit Employee Details	✕	✓	✓	✕	✕	✕	✕	✕
Manage Shift Rules	✕	✓	✓	✕	✕	✕	✕	✕
Configure Route Master	✕	✓	✕	✓	✕	✕	✕	✕
Add / Edit Vehicles	✕	✓	✕	✓	✕	✕	✕	✕
Add / Edit Drivers	✕	✓	✕	✓	✕	✕	✕	✕
Generate Trips (Auto)	✕	✓	✕	✓	✓	✕	✕	✕
Manual Trip Creation	✕	✓	✕	✓	✓	✕	✕	✕
View Live Tracking	✕	✓	✕	✓	✓	✕	Driver Own Trip	Employee Own Trip
Mark Trip Completed	✕	✕	✕	✕	✕	✕	✓	✓
Raise SOS	✕	✕	✕	✕	✕	✕	✓	✓

View Billing Reports	✕	✓	✕	✕	✕	✓	✕	✕
Generate Invoice	✕	✕	✕	✕	✕	✓	✕	✕
Export Reports	✕	✓	✕	✓	✕	✓	✕	✕
Access Audit Logs	✓	✕	✕	✕	✕	✕	✕	✕

---

1. Super Admin Panel – Subscription Tier Management

The platform is designed as a **multi-tenant SaaS system**, allowing multiple organizations to operate independently within the same application while maintaining strict data isolation.

The **Super Admin Panel** acts as the platform owner interface and provides centralized control over companies, subscription plans, system analytics, and operational monitoring.

Core Super Admin Functions

The Super Admin will have the following capabilities:

• Create and manage company accounts
• Manage subscription plans
• Monitor system usage across organizations
• Access platform analytics and system logs
• Control feature availability across subscription tiers

Configurable Subscription Tiers

The system supports configurable subscription tiers that allow the platform owner to offer different levels of functionality.

Example subscription tiers:

• Free
• Basic
• Premium
• Enterprise

Each tier can define:

• Feature access permissions
• Organization limits
• Module availability
• API usage limits

Feature Access Based on Subscription Plan

The Super Admin can configure which features are available in each subscription tier.

Example:

Feature	Free	Basic	Premium	Enterprise
Employee Management	✓	✓	✓	✓
Live Tracking	✓	✓	✓	✓
Route Optimization	✗	✓	✓	✓
Advanced Analytics	✗	✗	✓	✓

Vendor Marketplace	✗	✗	✓	✓

This allows the platform owner to restrict advanced capabilities to higher subscription plans.

Organization-Level Limits

Each subscription plan can define limits for organizations using the system.

Example configuration:

Plan	Employees	Vehicles	Trips per Day
Free	100	10	20
Basic	500	50	100
Premium	2000	200	Unlimited
Enterprise	Custom	Custom	Custom

These limits are enforced automatically by the backend services.

---

2. Reporting & Analytics Enhancements

The system already generates the following reports at the end of each month:

• Total trips
• Total distance traveled
• Driver performance
• Delay reports
• Cost per employee

To further enhance analytics capabilities, the following additional reports will be included.

Fleet Utilization Report

This report evaluates how efficiently vehicles are used across the transport network.

Metrics included:

• Number of trips per vehicle
• Average occupancy rate
• Idle vehicle percentage
• Total vehicle operating hours

Example:

Vehicle ID	Trips Completed	Utilization
V101	32	85%
V102	18	60%

This helps identify underutilized vehicles and optimize fleet usage.

Route Efficiency Report

This report analyzes route performance and helps improve route planning.

Metrics include:

• Total route distance
• Average trip duration
• Traffic delays
• Route deviation from optimal path

This report allows transport managers to identify inefficient routes and optimize travel time.

Route Consolidation Suggestions

The system includes an analytics engine that identifies opportunities to combine routes with low passenger occupancy.

Example:

Route A – 3 employees
Route B – 2 employees

System suggestion:

Merge routes to improve vehicle utilization and reduce operational cost.

---

3. Vendor Fleet Management (Multiple Transport Providers)

The system supports **multiple transport providers (vendors)**.

This is useful for organizations that outsource transportation to several fleet operators.

Vendor Management Features

The system will support:

• Vendor registration
• Vendor fleet management
• Vendor driver management
• Vendor contract tracking
• Vendor performance analytics

Each vendor can manage their own:

• Drivers
• Vehicles
• Assigned trips

This allows companies to manage transportation from multiple fleet vendors within a single platform.

---

4. Manual Trip Creation with Multiple Stops

Yes, the system supports **manual trip creation**.

Transport managers can manually create trips when required.

Features include:

• Assign vehicle
• Assign driver
• Add multiple pickup stops
• Add multiple drop locations
• Assign shift timing

Example trip route:

Stop 1 – Employee A
Stop 2 – Employee B
Stop 3 – Employee C

The system will automatically calculate:

• total route distance
• estimated travel time
• optimized stop sequence

---

5. Bulk Upload & Bulk Download

The system supports bulk data management to simplify large-scale operations.

Supported entities include:

• Employees
• Drivers
• Vehicles
• Routes
• Shift schedules

Supported formats:

• Excel (.xlsx)
• CSV (.csv)

Bulk Upload Workflow

1 Upload file
2 Validate records
3 Display error report (if any)
4 Import valid records
5 Generate summary report

The system will also allow **bulk data export** for reporting or external processing.

---

6. System Health Monitoring Dashboard

The platform includes a **system health monitoring dashboard** for administrators.

The dashboard will display:

• API response times
• Database performance metrics
• Server CPU and memory usage
• Active users and vehicles
• Background job status

Alerts will be triggered when:

• server load exceeds thresholds
• database latency increases
• service failures occur

This helps maintain platform reliability and operational visibility.

---

7. Edit and Delete Options for Forms

All system modules support full **CRUD operations (Create, Read, Update, Delete).**

Administrative users can:

• Create records
• Edit records
• Delete records
• Perform bulk deletion

Modules supporting CRUD operations include:

• Employees
• Drivers
• Vehicles
• Routes
• Trips
• Vendors

Both **single record operations** and **bulk operations** will be available.

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8. Database Backup and Data Recovery

The platform includes an automated **database backup and recovery system** to ensure data safety.

Backup Types

The system supports:

• Full database backup
• Organization-level backup
• Vendor-level backup

Backup Schedule

Example backup policy:

• Daily automated backup
• Weekly full snapshot
• Monthly archive backup

Backups will be stored in secure cloud storage such as:

• AWS S3

Data Restoration

The system will allow administrators to restore:

• full database
• specific organization data
• vendor-level data

Backup retention periods can be configured depending on organizational policies.